

ITEC 3100 – Introduction to Networks

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Lab 1B: *(Use a different color font for your written answers)*

Q1. What are the speed ranges for the 3 types of harddrives?

50 – 200 mb/s HDD

300-600 mb/s SSD

1500-7000 mb/s M.2

Q2. If the M.2 NVMe drive is so fast why is it not recommended on all systems?

Heat issues, cost, compatibility with older devices, and not really needed for everyday task

Q3. Explain what the difference between a LAN and WAN.

LAN is local WAN is Wide

Q4. List 5 different types of Servers and their purpose.

Web Server- Web content

DNS Server – Creates IP addresses

File Server- Stores and manages files for users

Database Server- Provides access to data

Cloud Server- information over the internet that can be delivered

Q5. Insert a picture of a Server Motherboard



Q6. Where are all these servers physically located?

Data centers

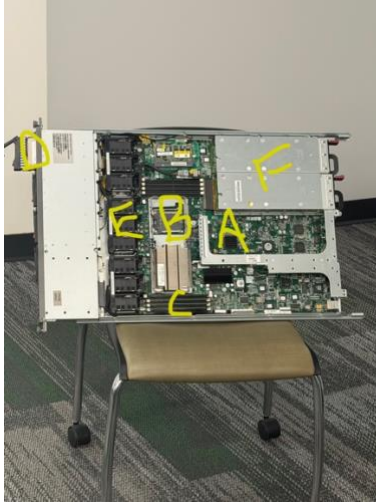
Q7. What is the purpose of the domain controller versus the authentication server?

The domain controller holds all usernames and passwords while the authentication server verifies identities

Q8. How are Servers related to the “cloud”?

Servers store everything physically while cloud stores everything virtually

Q9. Insert picture you took of the Server in class and label the following:



- a) Motherboard
- b) CPU
- c) RAM
- d) HDDs
- e) Fans
- f) Power supplies

Q10. Upload the screenshots we took in class.

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member: en1 flags=3<LEARNING,DISCOVER>
ifmaxaddr 0 port 8 priority 0 path cost 0
member: en2 flags=3<LEARNING,DISCOVER>
ifmaxaddr 0 port 9 priority 0 path cost 0
nd6 options=201<PERFORMNUD,DAD>
media: <unknown type>
status: inactive
ap1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=440<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
ether aa:23:9f:8a:c6:15
nd6 options=201<PERFORMNUD,DAD>
media: autoselect (none)
status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=440<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
ether 26:12:c3:f7:fe:82
inet6 fe80::94:fc11:a7af:50fa%en0 prefixlen 64 secured scopeid 0xb
inet 10.21.220.251 netmask 0xffff0000 broadcast 10.21.223.255
nd6 options=201<PERFORMNUD,DAD>
media: autoselect
status: active
awd10: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=440<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
ether 0a:7c:d2:9a:a2:5f
inet6 fe80::87c:d2ff:fe9a:a25f%awd10 prefixlen 64 scopeid 0xd
nd6 options=201<PERFORMNUD,DAD>
media: autoselect
status: active
utun0: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
inet6 fe80::845f:49d6:030d:b878%utun0 prefixlen 64 scopeid 0xe
nd6 options=201<PERFORMNUD,DAD>
llw0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether 0a:7c:d2:9a:a2:5f
inet6 fe80::87c:d2ff:fe9a:a25f%llw0 prefixlen 64 scopeid 0xf
nd6 options=201<PERFORMNUD,DAD>
media: autoselect (<unknown type>)
status: inactive
utun1: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
inet6 fe80::f9b:8da:f68f:8973%utun1 prefixlen 64 scopeid 0x10
nd6 options=201<PERFORMNUD,DAD>
utun2: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
inet6 fe80::e81:b1c:b2c:69e%utun2 prefixlen 64 scopeid 0x11
nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500
inet6 fe80::9c32:d894:d1e2:467c%utun3 prefixlen 64 scopeid 0x12
nd6 options=201<PERFORMNUD,DAD>
utun4: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
inet6 fe80::c4b:1bcb:865:3e9%utun4 prefixlen 64 scopeid 0x13
nd6 options=201<PERFORMNUD,DAD>
utun5: flags=8851<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
inet6 fe80::f8f:ec7b:77f2:348b%utun5 prefixlen 64 scopeid 0x14
nd6 options=201<PERFORMNUD,DAD>
shill@Ashilles-MacBook-Air ~ % ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=117 time=9.218 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=9.462 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=10.221 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=15.782 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=117 time=9.123 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=117 time=14.223 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=117 time=10.814 ms

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Last login: Tue Jan 20 14:26:15 on ttys000
shill@Ashilles-MacBook-Air ~ % ifconfig
lo0: flags=8849<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
options=120<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
inet 127.0.0.1 netmask 0xffff0000
inet6 ::1 prefixlen 128
inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anp10: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether 26:de:81:aa:ff:7b
media: none
status: inactive
anp11: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether 26:de:81:aa:ff:7c
media: none
status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether 26:de:81:aa:ff:5b
nd6 options=201<PERFORMNUD,DAD>
media: none
status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether 26:de:81:aa:ff:5e
nd6 options=201<PERFORMNUD,DAD>
media: none
status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
options=400<TSO4,TSO6,CHANNEL_IO>
ether 36:7b:d2:2c:16:40
media: autoselect <full-duplex>
status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
options=400<TSO4,TSO6,CHANNEL_IO>
ether 36:7b:d2:2c:16:44
media: autoselect <full-duplex>
status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=63<RXCSUM,TXCSUM,TSO4,TSO6>
ether 36:7b:d2:2c:16:40
Configurations:
id 0:0:0:0:0:0 priority 0 halldelay 0
maxage 0 holdont 0 proto stp maxaddr 100 timeout 1200
root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
lfilter disabled flags 0x0
member: en1 flags=3<LEARNING,DISCOVER>
ifmaxaddr 0 port 8 priority 0 path cost 0
member: en2 flags=3<LEARNING,DISCOVER>
ifmaxaddr 0 port 9 priority 0 path cost 0
nd6 options=201<PERFORMNUD,DAD>
media: <unknown type>
status: inactive
ap1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
options=440<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
ether aa:23:9f:8a:c6:15
nd6 options=201<PERFORMNUD,DAD>
media: autoselect (none)
status: inactive

```

Complete and upload to the same Chapter 1 Lab dropbox.

